Photolithography System

Equipment Technical Documentation (Sample)

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1. Equipment Overview

The Photolithography system is used to transfer circuit patterns onto semiconductor wafers using light exposure through a photomask. The system includes wafer alignment, exposure, and post-exposure bake processes.

Tool Type: Lithography
Model: LITHO-A300
Communication Protocol: SECS/GEM (HSMS)

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2. Process Parameters (Variables)

2.1 Status Variables (SV)

VID	Name	Type	Unit	Description
V1201	StagePosition_X	Float	mm	X-axis stage position
V1202	StagePosition_Y	Float	mm	Y-axis stage position
V1203	FocusOffset	Float	μm	Focus adjustment offset
V1204	ExposureIntensity	Float	mW/cm²	Exposure light intensity
V1205	AlignmentStatus	String	-	Alignment status of wafer
V1206	SystemStatus	String	-	Overall system status

2.2 Data Variables (DV)

VID	Name	Type	Unit	Description
V2201	RecipeID	String	-	Active lithography recipe
V2202	WaferID	String	-	Current wafer identifier
V2203	ExposureTime	Float	ms	Duration of exposure
V2204	OverlayError	Float	nm	Alignment error between layers
V2205	DoseValue	Float	mJ/cm²	Energy dose applied

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3. Equipment Events (CEID)

CEID	Event Name	Description	Linked VIDs
E301	WaferLoaded	Wafer loaded onto stage	V2202
E302	AlignmentComplete	Wafer alignment completed	V1201, V1202, V1205
E303	ExposureStart	Exposure process started	V1204, V2203
E304	ExposureEnd	Exposure process completed	V2203, V2205
E305	WaferUnloaded	Wafer removed from stage	V2202
E306	AlarmTriggered	Alarm condition detected	V1204, V1205

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4. Alarm Definitions

Alarm ID	Name	Severity	Description
A401	AlignmentFailure	Critical	Wafer alignment unsuccessful
A402	LowExposure	Warning	Exposure intensity below threshold
A403	FocusError	Critical	Focus offset exceeds limit
A404	StageMovementError	Critical	Stage positioning failure

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5. Remote Commands (RCMD)

Command	Description	Parameters
START	Start lithography process	RecipeID
STOP	Stop process	None
ALIGN	Perform wafer alignment	None
EXPOSE	Execute exposure	ExposureTime
RESET	Reset system after fault	None

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6. Equipment States

State ID	State Name	Description
S1	Idle	System ready
S2	Aligning	Wafer alignment in progress
S3	Exposing	Exposure process active
S4	Completed	Process completed
S5	Error	Fault condition

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7. State Transitions

- Idle → Aligning (Triggered by ALIGN command)
- Aligning → Exposing (Triggered by AlignmentComplete event)
- Exposing → Completed (Triggered by ExposureEnd event)
- Any State → Error (Triggered by AlarmTriggered event)
- Error → Idle (Triggered by RESET command)

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8. Communication Details

- Protocol: SECS/GEM
- Transport: HSMS (TCP/IP)
- Port: 5002

Example Messages:

- S1F1 → Establish communication
- S6F11 → Event report (AlignmentComplete, ExposureStart)
- S2F41 → Remote command (ALIGN, START, EXPOSE)

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9. Notes

- Accurate alignment is critical for multi-layer fabrication
- Exposure dose must be calibrated for each wafer batch
- System requires periodic calibration for optimal performance

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